

CS 690S Final Project Proposal

Learning Latent Intent Abstractions from Observing Others for Intent-Conditioned Reinforcement Learning

A central challenge in reinforcement learning is learning behavioral abstractions that support long-horizon reasoning, sparse-reward exploration, and transfer across related tasks. Standard RL typically relies on reward and trial-and-error interaction alone, while inverse reinforcement learning instead seeks to recover a reward function from demonstrations. I propose an intermediate, data-driven approach: learn a latent *intent* variable from observing other agents' trajectories, and use it as a reusable control abstraction for policy learning. Here, "intent" is not a hand-defined symbolic concept or a learned reward function, but a *latent temporally extended behavioral abstraction / control mode learned from trajectory segments*. This framing is inspired by recent intention-conditioned latent modeling in RL, especially Intention-Conditioned Flow Occupancy Models (InFOM), but my goal is not to reproduce that framework directly. Instead, I want to study whether observation-derived intent latents can become useful policy-conditioning variables in sparse-reward domains.

This project lies at the intersection of observational learning, theory-of-mind-style agent modeling, and hierarchical RL. Prior work on observational learning in RL shows that agents can benefit from watching other agents act in a shared environment rather than learning only from their own trial-and-error experience [1]. ToMnet similarly learns latent representations of other agents from trajectories and uses them to predict future behavior and hidden agent properties [2]. InFOM provides a more recent example of fully data-driven latent intention learning from heterogeneous trajectory data [3]. However, these lines of work do not directly answer the question I want to investigate: can an agent learn a reusable latent intent abstraction from observed trajectories that is useful for control, not just reward inference, next-action prediction, or agent characterization?

Problem. I will study whether an RL agent can learn a latent intent abstraction from observing other agents' trajectories, and whether this learned abstraction is more useful than state-only or trajectory-prediction-only baselines. The primary focus is representation learning: whether meaningful intent latents can be learned from behavior alone. The secondary focus is downstream benefit: whether conditioning a policy on the learned latent improves learning or transfer.

Experimental domain. The main domain will be a simple sparse-reward environment, likely a gridworld or MiniGrid-style setting, where agents exhibit distinct temporally extended behavior modes that are not reducible to a single state. Examples include different ways of approaching keys, doors, or reward-bearing objects. Montezuma's Revenge or a simplified Montezuma-like setting may be used only as motivation or as an optional extension.

Method. I will collect trajectories from other agents, likely scripted or pretrained policies with heterogeneous behavior. A trajectory encoder will infer a latent intent variable over trajectory segments. I will compare: (1) a plain RL baseline with no explicit latent intent, (2) a trajectory-model / agent-model baseline that encodes observed behavior but does not expose a reusable intent-conditioned control variable, and (3) the proposed method, where the acting policy conditions on the learned latent intent. If feasible, I may also include a lightweight state/subgoal abstraction baseline to test whether the learned latent captures more than ordinary state-based structure.

Hypothesis. Latent intent abstractions learned from observing others' trajectories will provide a more useful control representation than state-only or behavior-prediction-only baselines, especially in sparse-reward environments with temporally extended behavior modes. I expect the learned latent to organize trajectory segments into meaningful behavioral modes and, when used to condition policy, to improve sample efficiency or adaptation relative to plain RL.

Evaluation. The primary evaluation will measure representation quality: e.g., clustering or separability of trajectory segments, consistency of the inferred latent across segments from the same behavior mode, and prediction quality on held-out trajectories. The secondary evaluation will measure downstream control benefit: return, success rate, or sample efficiency when the learned latent is used by the policy. A positive result would show that observation can support learning reusable behavioral abstractions that are not hand-specified concepts, not reward functions, and not merely predictive agent embeddings.

Alignment relevance. This project is relevant to AI alignment because it studies how agents can acquire structured internal abstractions from observing others, rather than relying only on scalar reward and trial-and-error. In that sense, it is a small proof-of-concept for socially grounded representation learning that could eventually scale to richer multi-agent or agentic settings.

References

- [1] Bradly C. Stadie, Pieter Abbeel, and Ilya Sutskever. Third-Person Imitation Learning. *arXiv preprint arXiv:1703.01703*, 2017.
- [2] Neil C. Rabinowitz et al. Machine Theory of Mind. In *Proceedings of the 35th International Conference on Machine Learning*, 2018.
- [3] Chongyi Zheng, Seohong Park, Sergey Levine, and Benjamin Eysenbach. Intention-Conditioned Flow Occupancy Models. *arXiv preprint arXiv:2506.08902*, 2025.